

Income and Expense Record Keeping System

Descriptive Use Case

Use Case 01: Create User

Use Case	Create User
Primary Actor	User
Description	Creates a new account and stores the user's username, full name, email and password information so that the user can subsequently access the personal finance system.
Pre-condition	• The system is available. • The user is not already registered. • Required registration details are available.
Post-condition	• A new USER record is created. • The account becomes available for login. • The system confirms successful registration.
Basic Path	1. This use case starts when the user chooses to create an account. 2. The system displays the registration form. 3. The user enters a username, full name, email address and password. 4. The user submits the registration form. 5. The system validates that all required fields are present and that the username/email are unique. 6. The system creates the USER record with the supplied registration information. 7. The system confirms that registration was completed successfully.
Alternative Path	• If the user leaves a required field empty, the system identifies the missing information and asks the user to provide it. • If the username or email is already in use, the system rejects the registration and asks the user to choose different information.
Exceptional Path	• If the submitted registration data is invalid, the system does not create the account and displays an appropriate error message. • If the account cannot be stored because of a system/database failure, the system reports the failure and does not falsely confirm registration.

Use Case 02: Login

Use Case	Login
Primary Actor	User
Description	Authenticates a registered user and provides access to the appropriate personal finance dashboard.
Pre-condition	• A USER account exists. • The system is available. • Valid login credentials are available.
Post-condition	• The supplied credentials are verified. • An authenticated session is established. • The user's personal finance dashboard is displayed.
Basic Path	1. The use case starts when the user chooses Login. 2. The system displays the login interface. 3. The user enters a username/email and password. 4. The user submits the credentials. 5. The system verifies the supplied credentials against the registered account. 6. The system establishes an authenticated session. 7. The system displays the user's personal finance dashboard.
Alternative Path	• If the user submits incomplete credentials, the system asks the user to provide the missing information. • If the credentials are incorrect, the system denies access and allows the user to try again.
Exceptional Path	• If the authentication service is unavailable, the system reports that login cannot be completed at that time. • The system must not establish a session when the credentials cannot be successfully verified.

Use Case 03: Manage User Account

Use Case	Manage User Account
Primary Actor	User
Description	Allows an authenticated user to view and update permitted account information stored in the USER record.
Pre-condition	• The user is authenticated. • The USER record exists. • Account/profile settings are available.
Post-condition	• Valid account changes are stored in the USER record. • The user is informed that the update has been completed.
Basic Path	1. The user opens account/profile settings. 2. The system retrieves and displays the current account information. 3. The user reviews the displayed information. 4. The user edits permitted account fields. 5. The user submits the changes. 6. The system validates the updated information. 7. The system saves the valid changes to the USER record. 8. The system confirms successful completion.
Alternative Path	• If the user enters an invalid value, the system identifies the invalid field and requests correction. • If the submitted information conflicts with an existing account value, the system rejects the conflicting change and retains the previous valid information.
Exceptional Path	• If the account update cannot be stored, the system reports the failure and preserves the previous valid account information. • The system must not partially confirm an update that was not successfully saved.

Use Case 04: Manage Categories

Use Case	Manage Categories
Primary Actor	User
Description	Allows the user to create, view, update and delete categories used to classify income and expenses and support budget management.
Pre-condition	• The user is authenticated. • Category management is available.
Post-condition	• The requested category creation, update or deletion is completed. • The updated category information is stored and displayed.
Basic Path	1. The user opens Category Management. 2. The system displays the user's available categories. 3. The user selects create, edit or delete. 4. For create/edit, the user supplies or changes the category information. 5. The system validates the supplied category information. 6. The system stores the requested category change. 7. The system displays the updated category list.
Alternative Path	• If category information is incomplete or invalid, the system rejects the change and requests valid information. • If the user attempts to delete a category referenced by financial records, the system requires the dependent records to be handled before deletion.
Exceptional Path	• If the category cannot be stored or deleted because of a system/database failure, the system reports the failure and keeps the last valid category data. • The system must not indicate successful deletion when the category remains stored.

Use Case 05: Record Income

Use Case	Record Income
Primary Actor	User
Description	Records money received by the user, including amount, currency, date, description and category, and stores it as an INCOME record associated with the correct user.
Pre-condition	• The user is authenticated. • A suitable category exists. • Income details are available.
Post-condition	• A new INCOME record is stored. • The record is associated with the correct user and category. • A confirmation is displayed.
Basic Path	1. The user chooses Add Income. 2. The system displays the income entry form. 3. The user enters the amount, currency, date and description. 4. The user selects an appropriate category. 5. The user submits the income record. 6. The system validates the supplied income information. 7. The system classifies the transaction using the selected category. 8. The system stores the INCOME record. 9. The system displays a successful transaction confirmation.
Alternative Path	• If the amount, date, currency or category is missing or invalid, the system rejects the submission and identifies the information that must be corrected. • If the user selects an unavailable category, the system prevents the transaction from being saved until a valid category is selected.
Exceptional Path	• If storage fails after validation, the system reports the storage failure and does not display a false success message. • The system must not create a duplicate income record because of a failed submission.

Use Case 06: Manage Income

Use Case	Manage Income
Primary Actor	User
Description	Allows the user to view, edit or delete an existing income record belonging to the authenticated user.
Pre-condition	• The user is authenticated. • The selected INCOME record exists. • The selected record belongs to the user.
Post-condition	• The selected income record is successfully updated or deleted. • The resulting income history reflects the valid change.
Basic Path	1. The user selects an existing income record. 2. The system displays the user's income history. 3. The user selects the required income record. 4. The user chooses to edit or delete. 5. If editing, the user changes the permitted values. 6. The system validates the requested change. 7. The system updates or removes the INCOME record. 8. The system confirms the result.
Alternative Path	• If the user cancels the operation, the existing income record remains unchanged. • If an edited value is invalid, the system rejects the update and asks the user to correct it.
Exceptional Path	• If the record is unavailable or does not belong to the authenticated user, the system denies the operation and displays an error. • If the update/delete operation fails, the system preserves the valid record and reports the failure.

Use Case 07: Record Expense

Use Case	Record Expense
Primary Actor	User
Description	Records money spent by the user, including amount, currency, date, description, payment method and category, and stores it as an EXPENSE record.
Pre-condition	• The user is authenticated. • A suitable category exists. • Expense details are available.
Post-condition	• A new EXPENSE record is stored. • The record is associated with the correct user and category. • A confirmation is displayed.
Basic Path	1. The user chooses Add Expense. 2. The system displays the expense entry form. 3. The user enters amount, currency, date, description and payment method. 4. The user selects an appropriate category. 5. The user submits the expense. 6. The system validates the expense information. 7. The system classifies the transaction using the selected category. 8. The system stores the EXPENSE record. 9. The system displays a confirmation.
Alternative Path	• If required expense information is missing or invalid, the system rejects the submission and asks the user to correct it. • If the selected category is unavailable, the system prevents storage until a valid category is selected.
Exceptional Path	• If storage fails, the system reports the failure without falsely confirming the expense. • The system must avoid creating a duplicate expense when a transaction submission is unsuccessful.

Use Case 08: Manage Expense

Use Case	Manage Expense
Primary Actor	User
Description	Allows the user to view, edit or delete an existing expense record belonging to the authenticated user.
Pre-condition	• The user is authenticated. • The selected EXPENSE record exists. • The selected record belongs to the user.
Post-condition	• The expense record is successfully updated or deleted. • The user's expense history reflects the latest valid data.
Basic Path	1. The user selects an existing expense record. 2. The system displays the user's expense history. 3. The user selects a specific expense record. 4. The user chooses to edit or delete. 5. If editing, the user changes the required values. 6. The system validates the requested changes. 7. The system updates or removes the EXPENSE record. 8. The system confirms the result.
Alternative Path	• If the user cancels the operation, the original expense remains unchanged. • If an edited expense value is invalid, the system rejects the update and requests correction.
Exceptional Path	• If the record is unavailable, unauthorized or invalid, the system prevents the change and displays an error. • If the database operation fails, the system preserves valid expense data and reports the failure.

Use Case 09: Classify Transactions

Use Case	Classify Transactions
Primary Actor	User
Description	Associates an income or expense transaction with a CATEGORY so financial records can be organized and used for budget monitoring.
Pre-condition	• The user is authenticated. • The transaction exists or is being created. • A valid category exists.
Post-condition	• The transaction is successfully linked to the selected CATEGORY. • The classification is stored for future record viewing and budget monitoring.
Basic Path	1. The user creates or edits an income/expense transaction. 2. The system displays available categories. 3. The user selects a category. 4. The system verifies that the selected category is valid and available. 5. The system links the transaction to the selected CATEGORY. 6. The system saves the classification. 7. The transaction is now available under the selected category.
Alternative Path	• If the user changes the selected category before saving, the system uses the newly selected valid category. • If no category is selected, the system requests that the user choose a category.
Exceptional Path	• If the selected category is invalid or unavailable, the system rejects the classification. • If the classification cannot be saved, the system reports the failure and does not falsely indicate that the transaction was categorized.

Use Case 10: Create Budget

Use Case	Create Budget
Primary Actor	User
Description	Creates a budget limit for a selected category and applicable period so that spending can be monitored against the defined limit.
Pre-condition	• The user is authenticated. • The selected category exists. • The desired budget limit and period are known.
Post-condition	• A new BUDGET record is stored. • The budget is associated with the correct user and category. • A confirmation is displayed.
Basic Path	1. The user chooses Create Budget. 2. The system displays the available categories. 3. The user selects a category. 4. The user enters the budget limit and applicable period. 5. The user submits the budget. 6. The system validates the budget limit, period and category. 7. The system creates the BUDGET record. 8. The system confirms successful budget creation.
Alternative Path	• If the budget limit or period is invalid, the system rejects the budget and requests valid values. • If the selected category is unavailable, the system asks the user to select a valid category. • If conflicting budget information exists, the system flags the conflict instead of silently overwriting the existing information.
Exceptional Path	• If the budget cannot be stored, the system reports the failure and does not falsely confirm creation. • The system must preserve existing valid budget information when a new budget cannot be safely created.

Use Case 11: Manage Budget

Use Case	Manage Budget
Primary Actor	User
Description	Allows the user to view, edit or delete an existing budget belonging to the authenticated user.
Pre-condition	• The user is authenticated. • The selected BUDGET exists. • The selected budget belongs to the user.
Post-condition	• The budget is updated or deleted successfully. • Future budget monitoring uses the latest valid budget information.
Basic Path	1. The user selects an existing budget. 2. The system displays the available budgets. 3. The user selects a budget. 4. The user chooses to edit or delete. 5. If editing, the user changes permitted budget values. 6. The system validates the requested changes. 7. The system updates or removes the BUDGET record. 8. The system confirms the result.
Alternative Path	• If the user cancels the operation, the existing budget remains unchanged. • If an edited budget value is invalid, the system rejects the update and asks for correction.
Exceptional Path	• If the budget is unavailable or does not belong to the user, the system denies the operation. • If budget update/deletion fails, the system keeps the existing valid budget unchanged and reports the failure.

Use Case 12: Monitor Budget

Use Case	Monitor Budget
Primary Actor	User
Description	Calculates spending for a budget category and period and compares the spending with the budget limit to show the current budget status.
Pre-condition	• The user is authenticated. • The budget exists. • Relevant EXPENSE records can be retrieved.
Post-condition	• Spending for the selected period is calculated. • The budget limit is displayed. • The current status is shown as within, approaching or exceeding the limit where applicable. • The remaining or exceeded amount is calculated.
Basic Path	1. The user chooses to view a budget's status. 2. The user selects a budget. 3. The system retrieves the budget and category information. 4. The system retrieves relevant EXPENSE records for the category and period. 5. The system totals the relevant spending. 6. The system compares total spending with the budget limit. 7. The system calculates the remaining amount or amount exceeded. 8. The system displays the spending, limit and current budget status.
Alternative Path	• If there are no relevant expenses, the system treats spending as zero and displays the resulting budget status. • If spending is close to the budget limit, the system displays the applicable approaching status.
Exceptional Path	• If the budget or expense data cannot be retrieved, the system reports that the status cannot be calculated. • The system must not display an unreliable budget status when required data is unavailable.

Use Case 13: View Financial Records

Use Case	View Financial Records
Primary Actor	User
Description	Displays the authenticated user's stored income and expense history for review and allows available filtering by fields such as date, category or transaction type.
Pre-condition	• The user is authenticated. • The user has permission to view their own records.
Post-condition	• The requested financial records are retrieved and displayed. • The user can review and, where supported, filter the displayed records.
Basic Path	1. The user chooses Financial Records. 2. The system identifies the authenticated user. 3. The system retrieves the user's INCOME and EXPENSE records. 4. The system displays the transaction information. 5. The user reviews the financial records. 6. The user may apply available filters such as date, category or transaction type. 7. The system displays the records matching the selected filters.
Alternative Path	• If no records match the selected filters, the system displays an empty-result message. • If the user changes or removes a filter, the system refreshes the displayed records using the new criteria.
Exceptional Path	• If record retrieval fails, the system displays an error instead of showing incomplete information as if it were complete. • The system must not expose another user's income or expense records.

Use Case 14: Logout

Use Case	Logout
Primary Actor	User
Description	Ends the authenticated session and prevents continued access through that session, returning the user to the Login interface.
Pre-condition	• The user is authenticated. • An active session exists.
Post-condition	• The authenticated session is terminated or invalidated. • Protected access through the session is ended. • The user is returned to Login.
Basic Path	1. The user selects Logout. 2. The system receives the logout request. 3. The system terminates or invalidates the active session. 4. The system ends access to protected pages through the session. 5. The system redirects the user to the Login page.
Alternative Path	• If the session has already expired, the system redirects the user to Login and does not treat the expired session as authenticated.
Exceptional Path	• If session termination fails, the system displays an error instead of falsely confirming logout. • The system should prevent continued protected access whenever the session can no longer be considered valid.